

DATE: June 9, 2021

TO: Mauricio Leon, Research

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SUBJECT: Met Council forecasts extended to housing and land use: for BAU scenario

This memo is documentation of how Met Council's local forecast database, with socioeconomic forecast by community and TAZ, is extended to provide projections of 2018-2040 housing stock by housing type and projected 2018-2040 land use acreages.

City and town forecasts and TAZ allocations of population, households, and employment were initially allocated using the CubeLand model in 2014-15. A short methodology explanation can be found online: [https://metro council.org/Data-and-Maps/Publications-And-Resources/Files-and-reports/Forecast-Methodology-Report-\(October-2019\).aspx](https://metro council.org/Data-and-Maps/Publications-And-Resources/Files-and-reports/Forecast-Methodology-Report-(October-2019).aspx) City and town forecasts were adjusted, post-model, to reflect known development plans or land policy constraints; the adjustments were approved by Met Council in July 2015.

During the comprehensive plan update cycle, 2016-2020, local governments requested or negotiated local forecast revisions in some cases. Met Council requires the revised forecasts must be consistent with land supply, development plans or expectations, and regional system plans. Where those conditions were met, all forecast change requests were approved. Also during comprehensive plan updates, local governments were invited to review and modify the subcity, spatial pattern of expected growth, allocating forecasts to subcity Transportation Analysis Zones. By December 2020, approximately 90% of zonal allocations are from local governments themselves; the remaining balance of zonal allocations are projected by Met Council staff in lieu of local governments' plans.

The current vintage (currently in our GIS Data Catalog) allocations of population, households, and employment by TAZ are a starting point for the projection of future land use and future housing stock. Data is available here: <https://gisdata.mn.gov/dataset?tags=forecasts&q=taz+forecast>

Starting point nonresidential land use (2016) comes from Met Council's 2016 generalized land use inventory. Starting point residential land use and housing counts(2018) come from analysis of 2018 property parcels. Both are parsed or summarized by TAZ.

Projected housing growth by TAZ is fitted to match each TAZ's households growth after 2018. Housing stock growth shares for SFD and MultiFamily are proportionate to each type's share of planned housing capacity within the TAZ. (At this time, townhomes, duplexes, triplexes, etc., are grouped with MultiFamily.) Capacity is calculated from acreage and allowed housing density per local plans. If the housing growth is negative (a decline), then housing growth shares are instead proportionate to the existing housing stock, rather than the planned capacity. (Note: Conceptually, local households declines should be rare. But in the current comprehensive plan update cycle, we have invited local planners to make the decision of how forecasts are allocated within their jurisdiction -- and some have provided local TAZ allocations that include small losses of households.)

Projected residential land use growth for SFD and MultiFamily is calculated as: the housing growth for each multiplied by a land consumption rate for each type, specific to the TAZ. If the housing growth is positive, then the land consumption

rate is 1 divided by allowed housing density per local plans. However, the cumulative total is capped; the land use growth is not allowed to exceed the available land supply. If the housing growth is negative (a decline), then the (negative) land consumption rate is instead 1 divided by observed, existing housing density.

Projected nonresidential land use growth by TAZ is similarly fitted to match each TAZ's employment growth after 2018, and parsed by real estate submarket: industrial, institutional, commercial. (At this time, office towers are grouped with commercial.) Nonresidential growth in a local area is difficult to project with accuracy. For expediency then, the BAU scenario expects, total nonresidential land use growth is calculated as: the employment growth multiplied by a land consumption rate, specific to the TAZ. The land consumption rate is 1 divided by observed, existing jobs density. However, the cumulative total is capped; the land use growth is not allowed to exceed the available land supply.

The three submarkets of nonresidential land use will each grow proportionate to their local zone share of available, planned land supply acreage. If the employment and nonresidential growth is negative (a decline), then nonresidential growth shares are instead proportionate to the existing land use, rather than the planned capacity.

Available, planned land supply, discussed in the section above, is land that is guided for future use (i.e. development is allowed) and is (a) presently undeveloped (undeveloped and also farmland) or (b) presently underdeveloped. Our indicator of underdevelopment is a low building-value/land-value ratio (in the range 0 to 0.25). With this definition, there are fully-developed TAZs where the available, planned land supply is minimal or zero. In those cases, land consumption rates may be set at or near zero, meaning that the forecasted households growth or employment growth will be redevelopment and intensification on the existing land use footprint.

The resulting database.

The resulting database is stored on the Research shared drive:

N:\CommDev\Research\Research\Todd\FORECAST_LU_TAZCTU.xlsx

The first 3 columns indicate that each record is the combination of TAZ, county, and city/town. The table can be collapsed (with pivot table) to obtain city/town totals.

The next 24 columns are our published forecasts by TAZ table; you've seen those columns before.

The right side of the table is all new info.

- SFD_Units calculated for 2018, and projections for future years.
- MF_Units for 2018; and projections for future years.
- MH_Units is mobile home parks counts from 2018 parcels data.
- SFD_Acres of SFD land projected for 2018-2040.
- MF_Acres of MF land projected for 2018-2040.
- MH_Acres is mobile home parks land in 2018; and we do not have a good way to calculate change there, so have treated this as an unchanging constant.
- Res_Acres is the summation of SFD_Acres + MF_Acres + MH_Acres.
- IND_Acres of Industrial land projected for 2018-2040.
- INS_Acres of Institutional land projected for 2018-2040.
- COM_Acres of Commercial land projected for 2018-2040.
- NonRes_Acres is the summation of NonResidential types.